



Structural Segmentation with Transformers: SpecTNT for Functional Music Structure Analysis - Extension 2

Fharook Shaik , Douglas W. Cunningham, Prashant Varadarajan
Department of Graphical Systems, Univ. BTU Cottbus-Senftenberg, Cottbus, Germany

1. INTRODUCTION

Music Structure Analysis (MSA) traditionally divides songs into segments with abstract labels (A/B/C) or focuses only on chorus/boundary detection. Explicit semantic labelling (e.g., “verse”, “chorus”, “bridge”) enables powerful applications such as structure-based navigation, automated remixing, live visual synchronization, and real-time analysis of song fragments (e.g., TikTok Clips).

Deep-learning approaches like SpecTNT (Spectral-Temporal Transformer in Transformer) achieve strong results by predicting functional labels directly from audio [1]. However, frame-wise predictions often suffer from fragmentation and temporal inconsistency. This project reimplements the SpecTNT baseline and extends it with a lightweight temporal smoothing decoder to improve section coherence without altering the model architecture.

2. RESEARCH QUESTION

Following Wang et al. (2022) [1], this project performs functional structure analysis using a 7-class taxonomy: *intro, verse, chorus, bridge, instrumental, outro, silence*.

How does applying a temporal smoothing decoder (Viterbi algorithm or a transition-penalty smoothing) as post-processing on top of SpecTNT’s frame-wise predictions affect overall accuracy and temporal consistency of the predicted musical sections?

Hypothesis: Because the sequence of sections in popular music follows regular patterns, explicitly smoothing the frame-wise probabilities during post-processing will significantly improve structural segmentation metrics (HR.5F, PWF, Sf) and macro F1 scores without requiring modifications to the underlying neural network architecture.

3. METHODOLOGY

- **Dataset:** *Harmonix Set* [2] - 912 western pop songs with functional segment annotations.
- **Baseline:** *Reimplementation of the SpecTNT model* [1].
- **Extension:** *SpecTNT + Temporal Smoothing Decoder (Viterbi algorithm or transition-penalty smoothing) as post-processing step.*
- **Evaluation Metrics:** *Hit Rate (HR.5F), Frame-wise Accuracy (ACC), Pair-Wise F-measure (PWF), Normalized Entropy Score (Sf), Macro F1 (7 classes), Chorus Hit Rate (CHR.5F).*

